Norges miljø- og biovitenskapelige universitet

TEL200 – Introduction to Robotics

David A. Anisi

Chapter 2: Representing position and orientation

ABB Fanta Challenge – What is needed?

ABB Fanta Challenge – What is needed?

- Know position and orientation (Pose) — Ch. 2
  - Coordinate systems and representations
  - Not only in 2D & 3D but also in joint- or configuration space
- Not only ***now*** but also as a function of time: Describe motion & time — Ch. 3
  - Time varying pose, paths, trajectories
  - Accelerating bodies and coordinate systems
  - Dynamics of rigid, mechanical bodies
  - Creating smooth paths and trajectories
- Kinematic modelling & control of robot (arm) — Ch. 7-8

Outline

- Ch. 2.1 Foundations
- Ch. 2.2: Working in Two Dimensions (2D)
- Ch 2.3: Working in Three Dimensions (3D)
- Ch 2.4.9 Configuration space

chap2.ipynb

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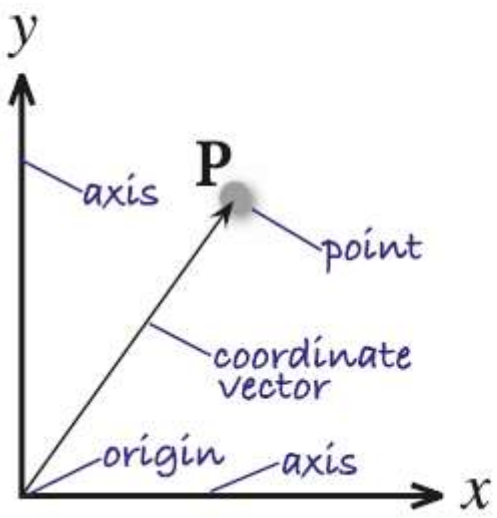
Introduction

- A point **P** in space is a familiar concept from mathematics and can be described by a ***coordinate vector*** (Fig. a.)
- If we attach a ***coordinate frame*** to an object, as shown in Fig. b, we can describe every point within the object as a constant vector with respect to that frame.
- What is the main benefit of this?
- Now we can describe the ***position*** and ***orientation***[1] – the pose – of that coordinate frame $\{B\}$ with respect to the reference coordinate frame {0} as ξ_B.

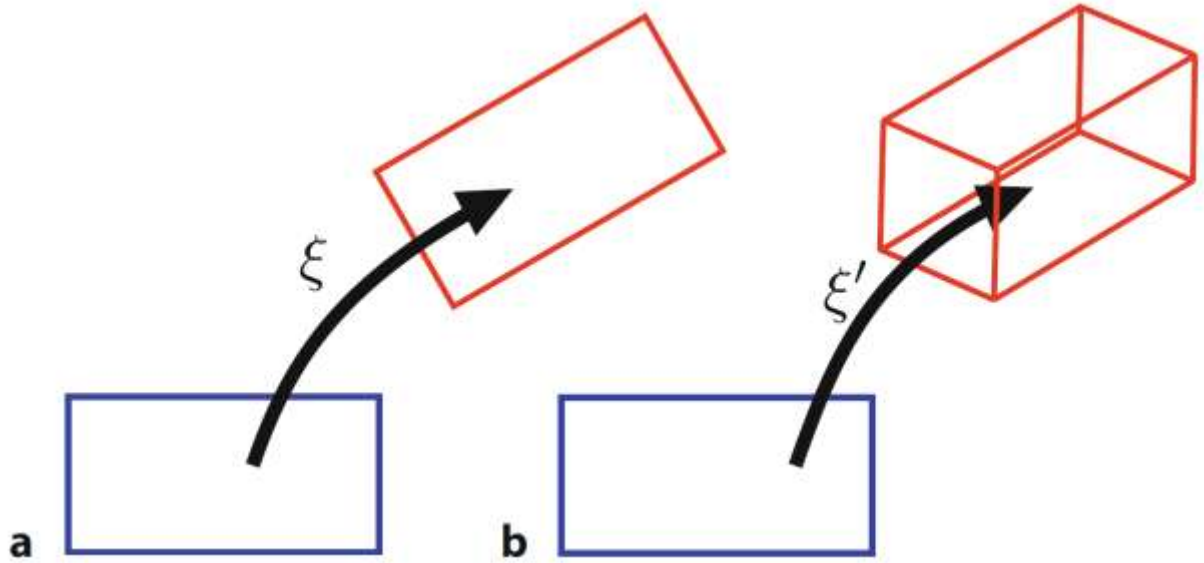
a



[1] Attitude is an alternative nomenclature for Orientation

Relative pose

- Blue and red objects have different pose: ***position*** and ***orientation***
- Rigid motion (no change in shape) consisting of ***translation*** and ***rotation***
- Motion, ξ * is always defined with respect to an initial pose.



Rigid motion in 2D (a) and 3D (b)

* Pronounced ksi

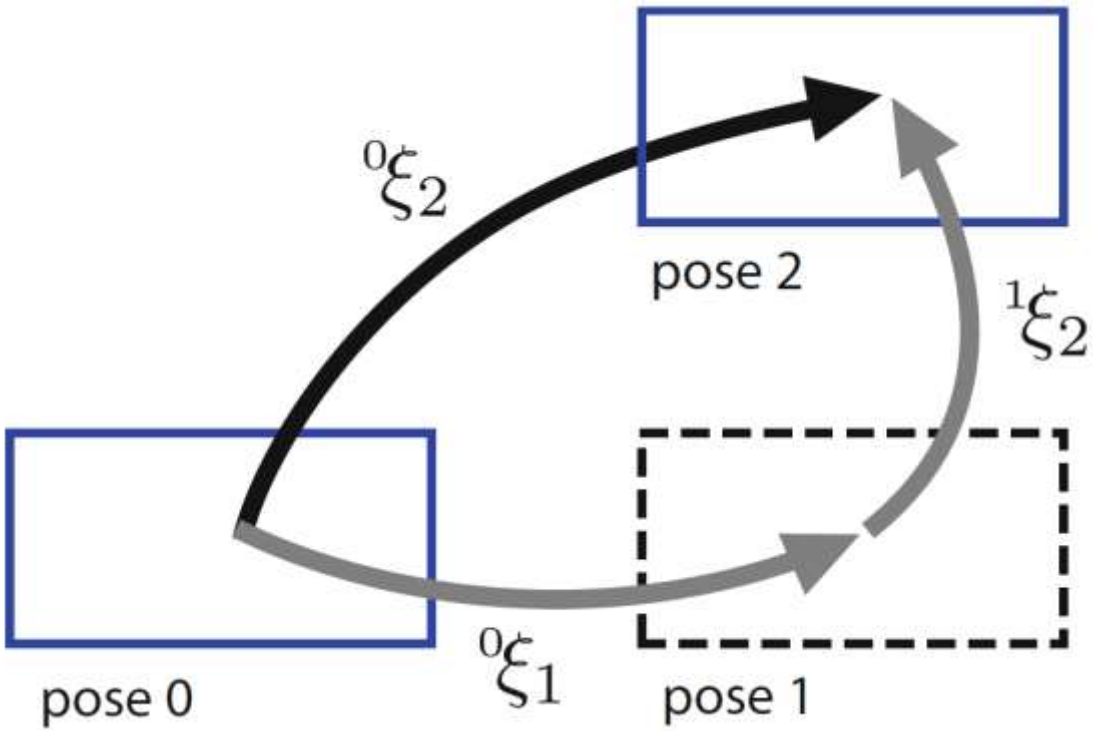
Relative pose

- Explicitly indicating start- (X) and end-pose (Y): ${}^{X}\xi_{Y}$
- Motion composition: ${}^{0}\xi_{2} = {}^{0}\xi_{1} \oplus {}^{1}\xi_{2}$
  - is **not** commutative
  - has an inverse or opposite motion

$$ {}^{X}\xi_{Y} \oplus {}^{Y}\xi_{X} = {}^{X}\xi_{X} = \varnothing $$

Introducing: ${}^{Y}\xi_{X} \equiv \ominus\, {}^{X}\xi_{Y}$

We can write: ${}^{X}\xi_{Y} \ominus {}^{X}\xi_{Y} = \varnothing$

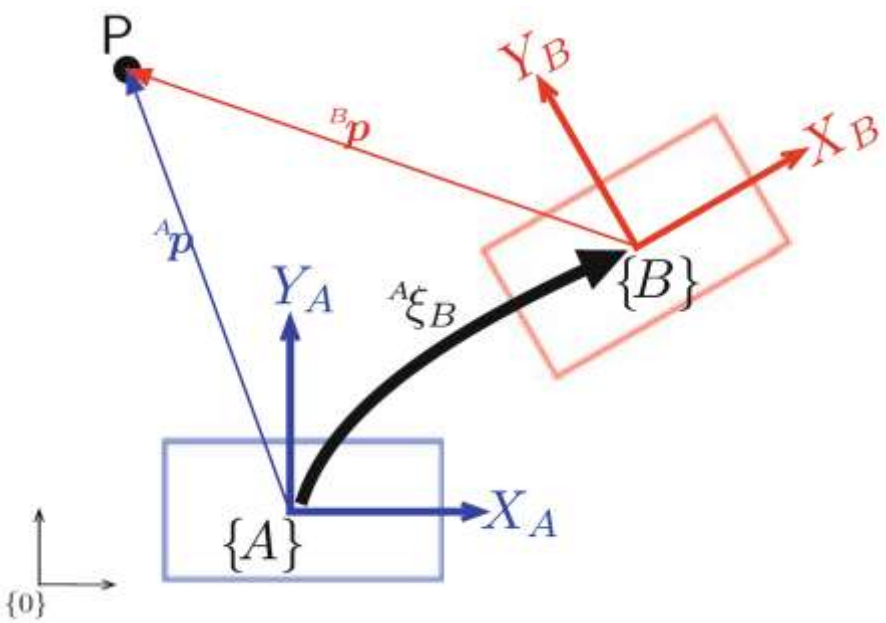


Relative pose

- The point **P** can be described with respect to *either* coordinate frame by the vectors ${}^{A}p$ or ${}^{B}p$ respectively

$$ {}^{A}\boldsymbol{p} = {}^{A}\xi_{B} \cdot {}^{B}\boldsymbol{p} $$

- ${}^{A}\xi_{B}$: The relative pose of {B} to {A}
- Two interpretations for ${}^{A}\xi_{B}$:
  - Picking up A and transforming it with ${}^{A}\xi_{B}$ leaves A in B's place
  - Let ${}^{B}p$ be the coordinates of ***p*** in {B}. Transforming ${}^{B}p$ with ${}^{A}\xi_{B}$ gives coordinates of ***p*** in A, denoted ${}^{A}p$



Relative pose

- An important characteristic of relative poses is that they can be *composed* or *compounded*.

$$ {}^{A}\xi_{C} = {}^{A}\xi_{B} \oplus {}^{B}\xi_{C} $$

- The $\oplus$ operator denotes compounding of relative poses
- Cancel out intermediate subscripts and superscripts

$$ {}^{X}\xi_{Z} = {}^{X}\xi_{\cancel{Y}} \oplus {}^{\cancel{Y}}\xi_{Z} $$

- The point **P** can e.g., be described by

$$ {}^{A}\boldsymbol{p} = \left({}^{A}\xi_{B} \oplus {}^{B}\xi_{C} \right) \cdot {}^{C}\boldsymbol{p} $$

- Q: How is this relevant and important for robotics?

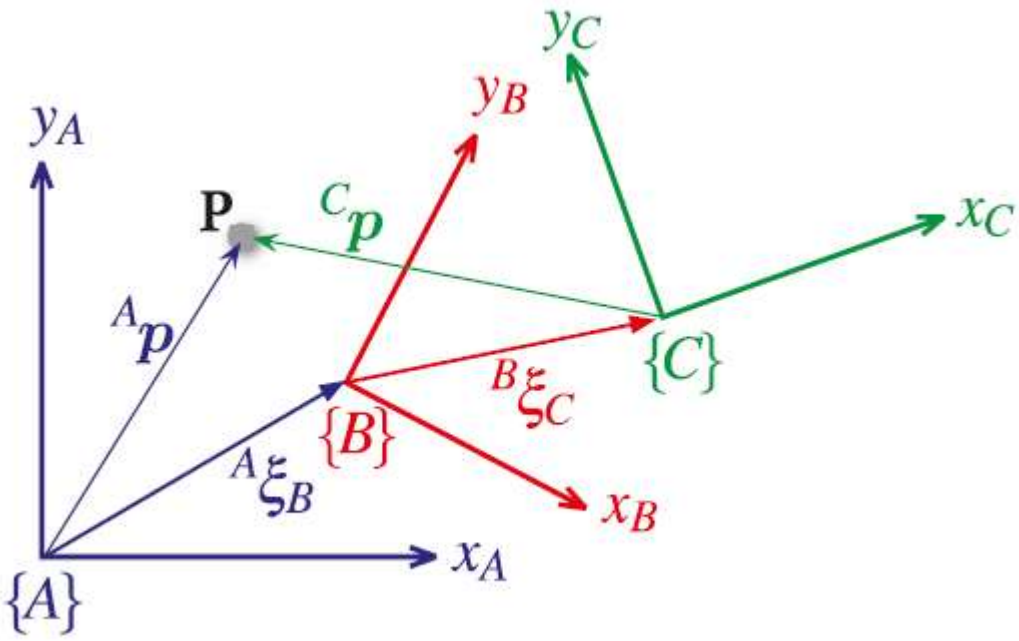
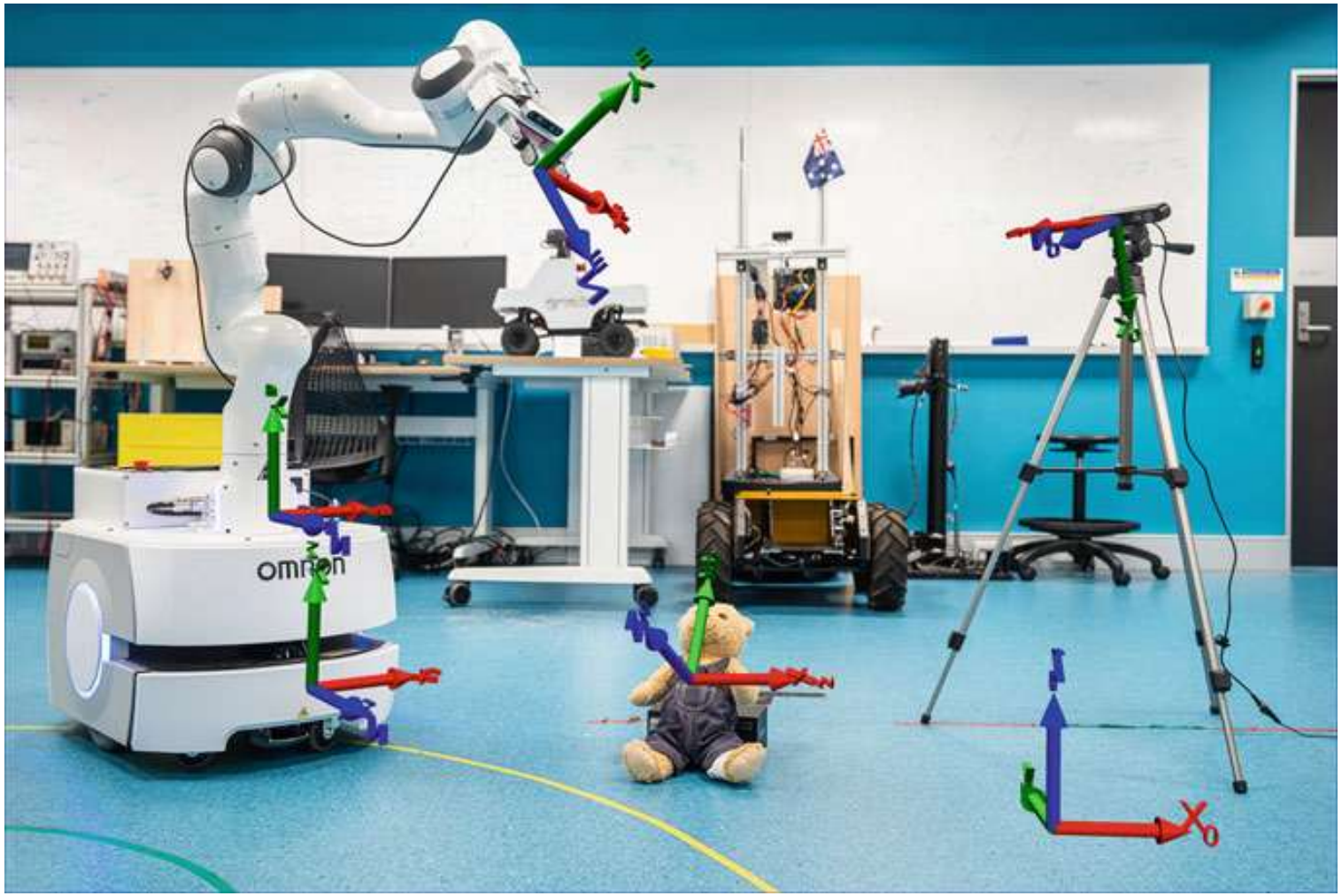


Fig. 2.3.
The point P can be described by coordinate vectors relative to either frame $\{A\}$, $\{B\}$ or $\{C\}$. The frames are described by relative poses

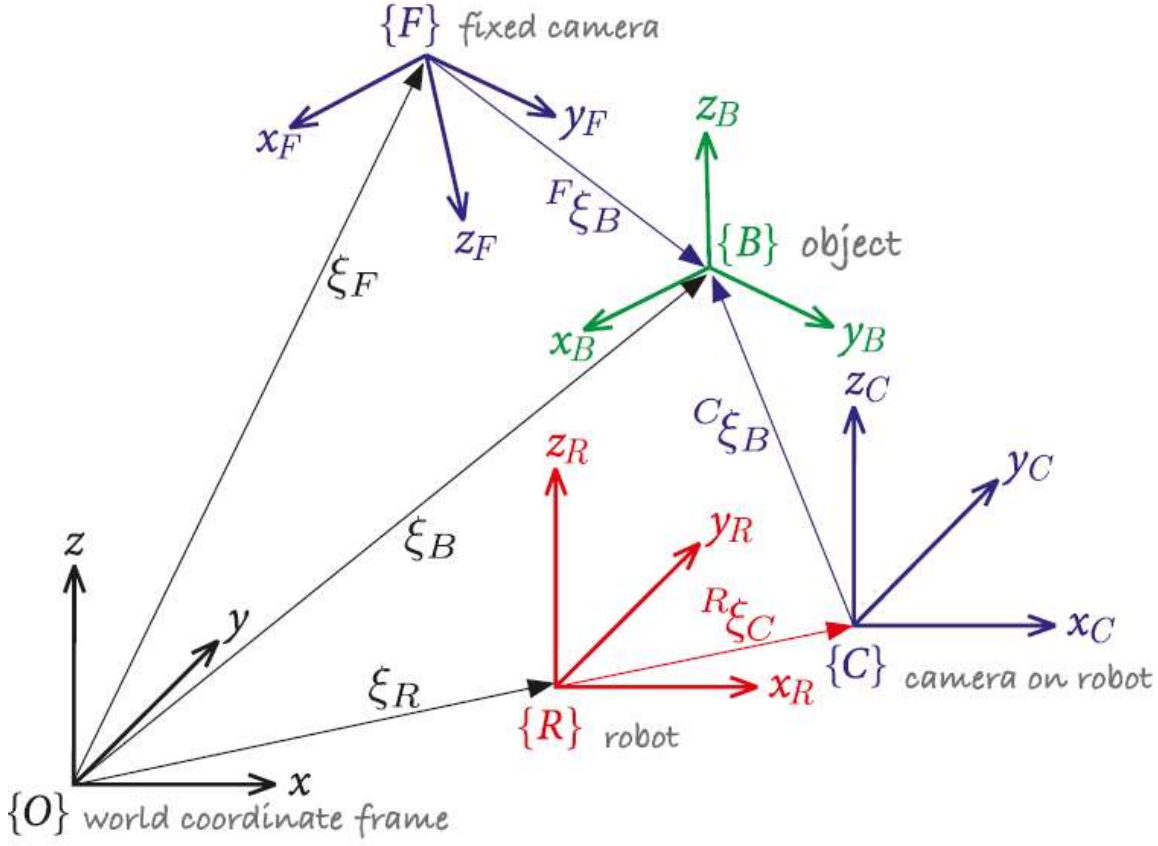
Examples of robot coordinate frames in robotics



Introduction

- Figure 2.4 shows a 3-dimensional example in a graphical form where we have attached 3D coordinate frames to the various entities and indicated some relative poses.
- Q: What is the pose of the robot relative to the fixed camera?

Fig. 2.4.
Multiple 3-dimensional coordinate frames and relative poses

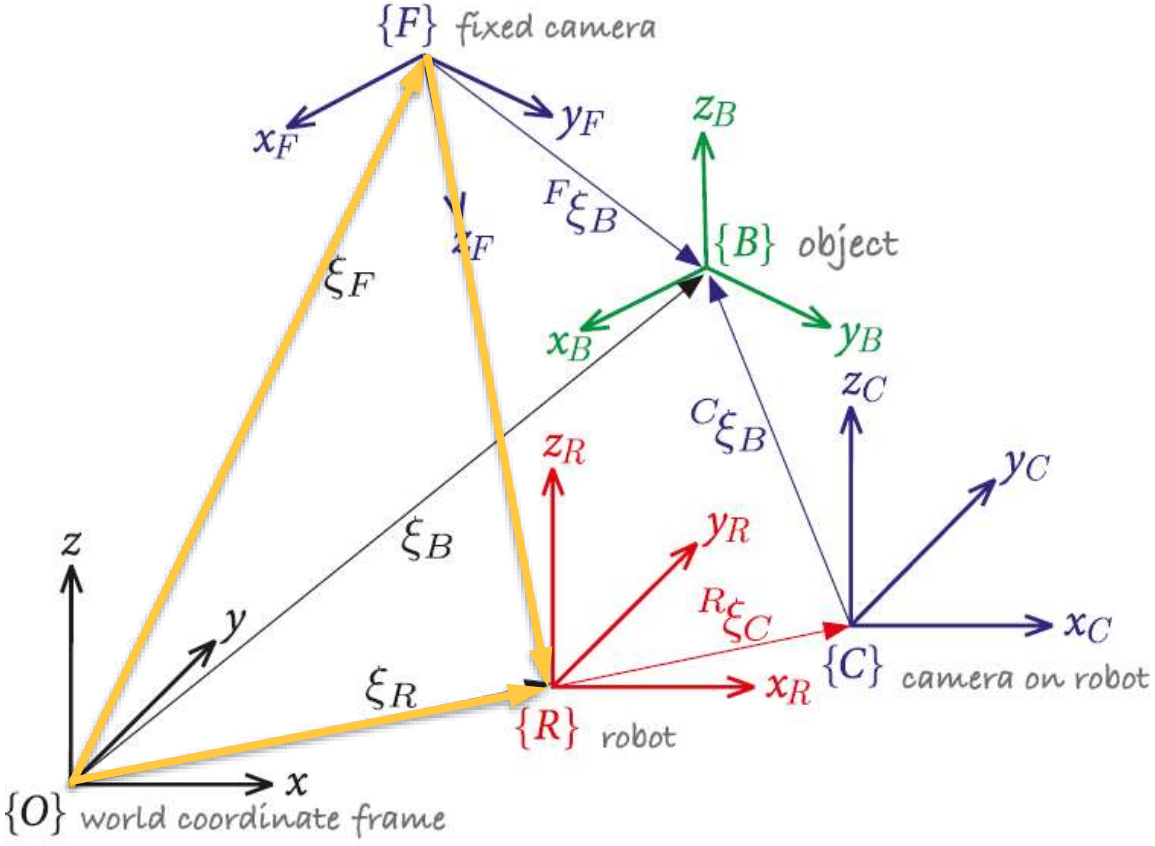


Introduction

- Recalling that we can compose relative poses using the $\oplus$ operator we can write some spatial relationships:

$$\xi_F \oplus {}^F\xi_R = \xi_R \tag{2.4}$$

Fig. 2.4.
Multiple 3-dimensional coordinate frames and relative poses



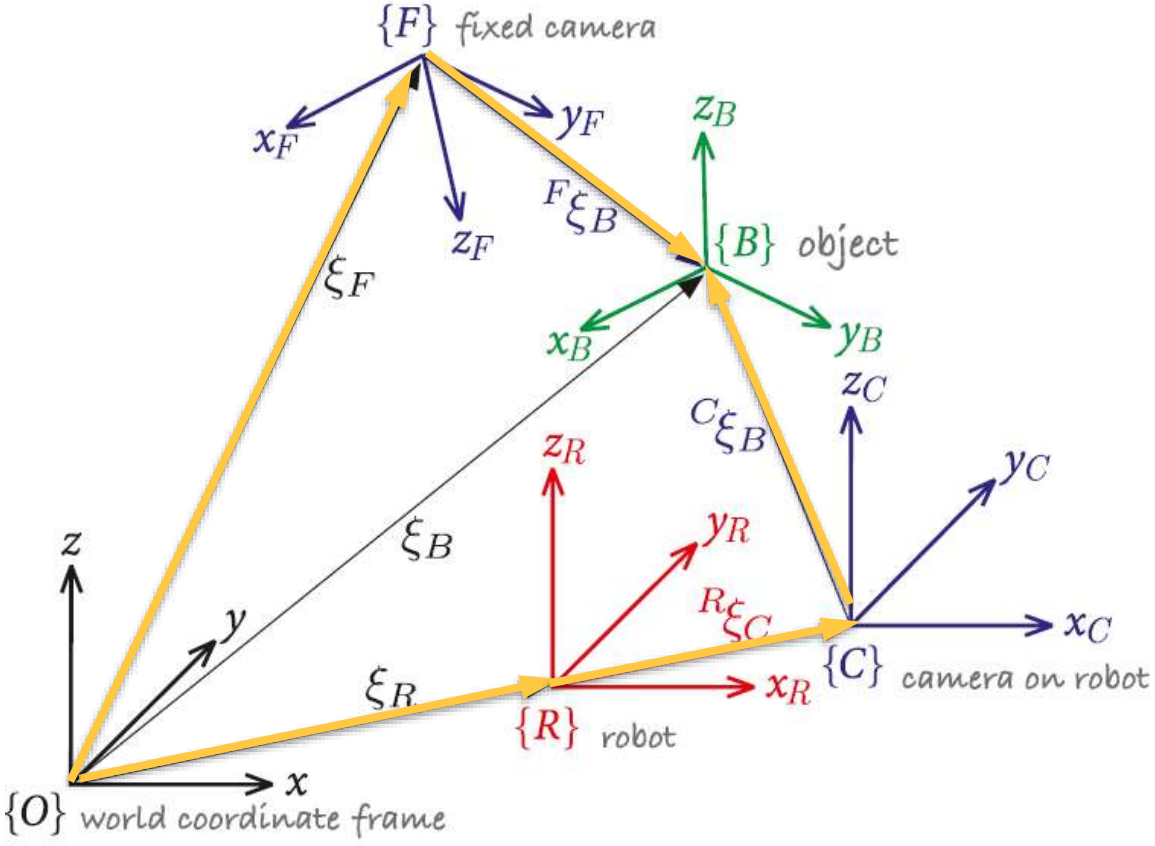
Introduction

- Recalling that we can compose relative poses using the $\oplus$ operator we can write some spatial relationships:

$$\xi_F \oplus {}^F\xi_R = \xi_R \tag{2.4}$$

$$\xi_F \oplus {}^F\xi_B = \xi_R \oplus {}^R\xi_C \oplus {}^C\xi_B \tag{2.5}$$

Fig. 2.4.
Multiple 3-dimensional coordinate frames and relative poses



Introduction - Remark

- **Algebraic rules** for poses and relative poses

$$\xi \oplus 0 = \xi, \quad \xi \ominus 0 = \xi \tag{R.1}$$

$$\xi \ominus \xi = 0, \quad \ominus \xi \oplus \xi = 0 \tag{R.2}$$

where 0 represents a zero relative pose.

- Inverse of a pose:

$$\ominus {}^{X}\xi_{Y} = {}^{Y}\xi_{X} \tag{R.3}$$

- Relative pose composition

$${}^{X}\xi_{Y} \oplus {}^{Y}\xi_{Z} = {}^{X}\xi_{Z} \tag{R.4}$$

Introduction - Remark

- It is important to note that the algebraic rules for poses are different to normal algebra and that composition is **not** commutative

$$\xi_1 \oplus \xi_2 \neq \xi_2 \oplus \xi_1 \tag{R.5}$$

except when

$$\xi_1 \oplus \xi_2 = 0.$$

Introduction

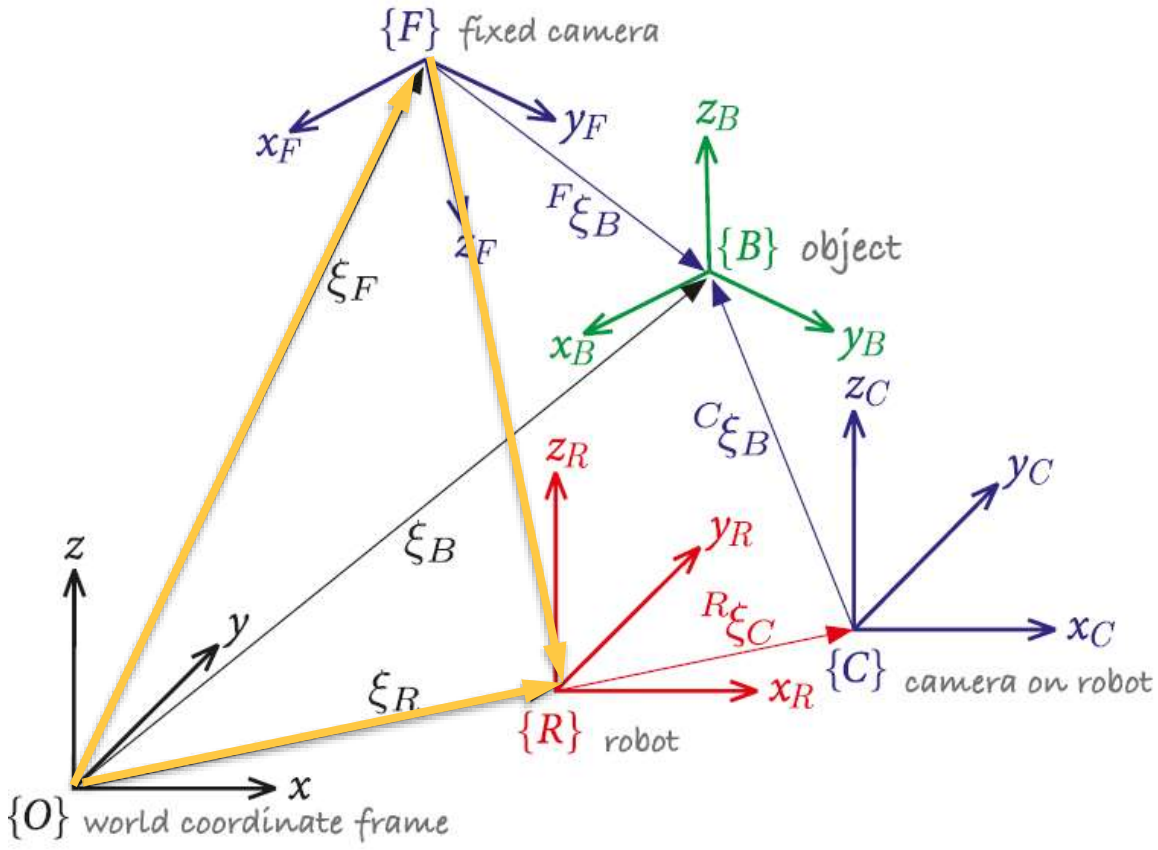
- Q: What is the pose of the robot relative to the fixed camera?

$$\xi_F \oplus {}^F\xi_R = \xi_R \tag{2.4}$$

$$\cancel{\ominus\xi_F} \oplus \cancel{\xi_F} \oplus {}^F\xi_R = \ominus\xi_F \oplus \xi_R \tag{2.6}$$

$${}^F\xi_R = \ominus\xi_F \oplus \xi_R \tag{2.7}$$

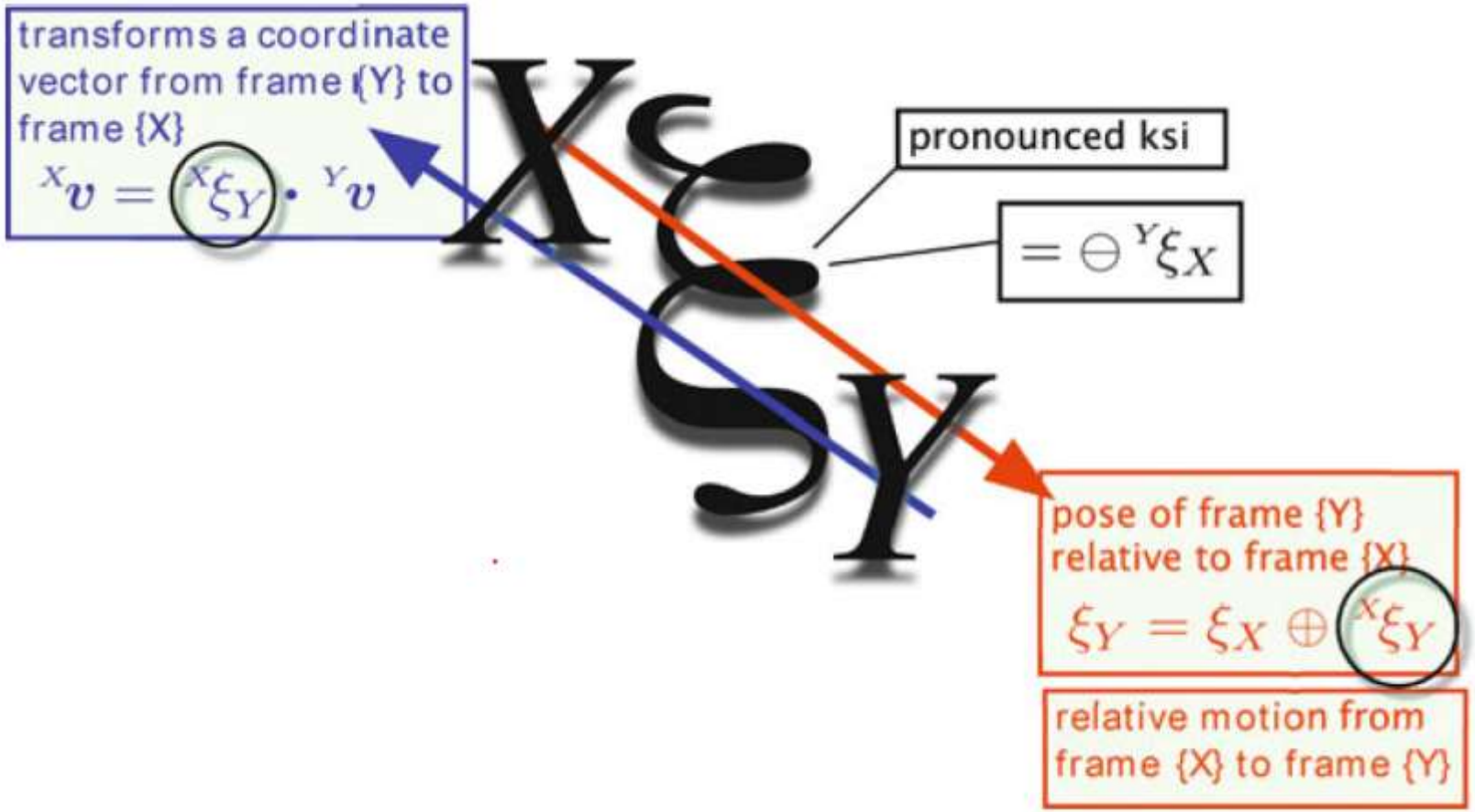
Fig. 2.4.
Multiple 3-dimensional coordinate frames and relative poses



Foundations - Recap

1) A **point** is described by a coordinate vector that represents its displacement from the origin of a reference coordinate system;
2) A **set of points** representing a rigid object can be described by a single coordinate frame, and the constituent points, described by displacements from it
3) The position and orientation of an object's coordinate frame is referred to as its **pose**;
4) A **relative pose** describes the pose of one coordinate frame with respect to another and is denoted by an algebraic variable ξ;
5) A coordinate vector describing a point can be represented with respect to a different coordinate frame by applying the relative pose to the vector;
6) We can perform algebraic manipulation of expressions written in terms of relative poses and the operators $\oplus$ and $\ominus$ and the concept of a null motion $\varnothing$;

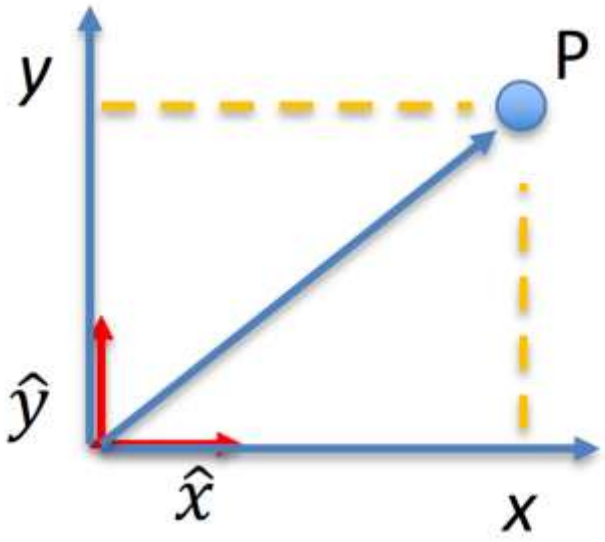
Summary - A key takeaway about pose



Working in Two Dimensions (2D)

- A point **P** is represented by its x- and y-coordinates (x, y) and unit-vectors parallel to the axes are denoted $\hat{\boldsymbol{x}}$ and $\hat{\boldsymbol{y}}$.

$$\boldsymbol{p} = x\hat{\boldsymbol{x}} + y\hat{\boldsymbol{y}} \tag{2.6}$$



Working in Two Dimensions (2D)

- Figure 2.6 shows that the origin of $\{B\}$ has been displaced by the vector $t = (x, y)$ and then rotated counter-clockwise by an angle θ.
- We can use ${}^{A}\xi_{B} \sim (x, y, \theta)$ to represent this relative pose
- But is this a convenient representation?

$$\cancel{(x1, y1, \theta1) \oplus (x2, y2, \theta2)}$$

- We decompose the problem into two parts: Rotation and Translation

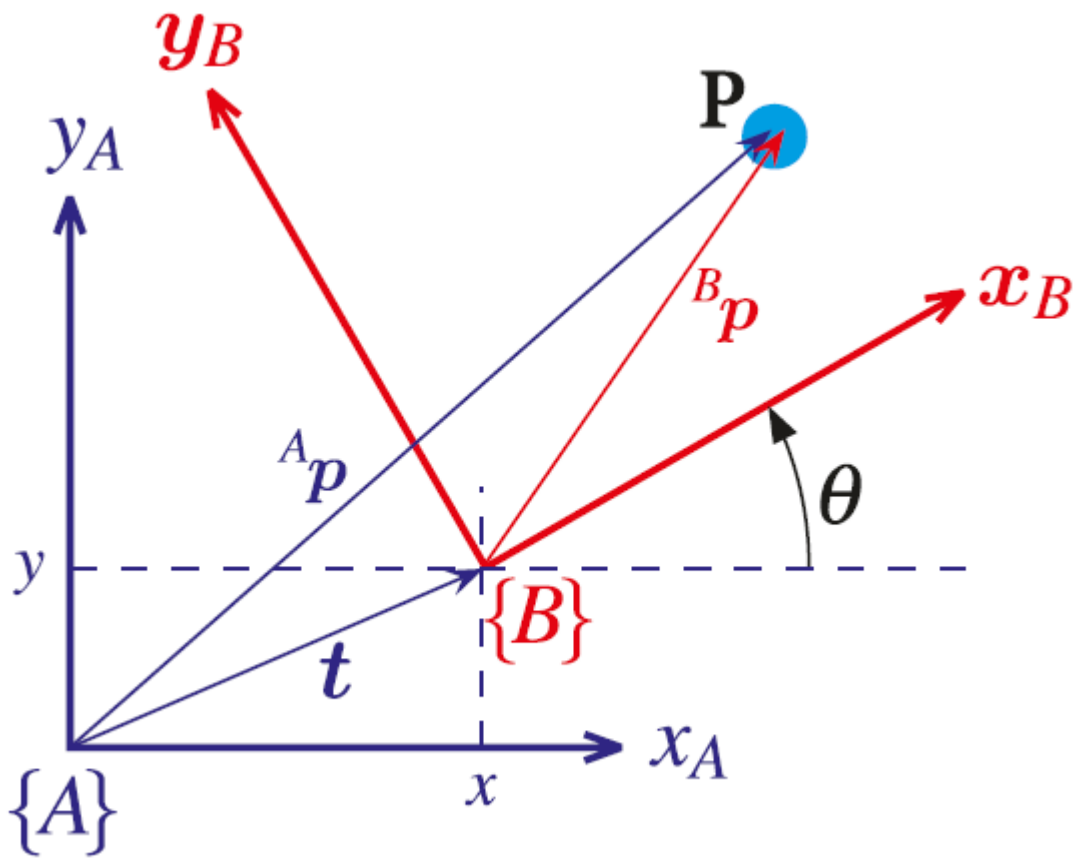


Fig. 2.6.
Two 2D coordinate frames $\{A\}$ and $\{B\}$ and a world point P. $\{B\}$ is rotated and translated with respect to $\{A\}$

Rotation in Two Dimensions (2D)

- Frame $\{A\}$ is completely described by $\hat{x}_A$ and $\hat{y}_A$
- Frame $\{B\}$ is completely described by $\hat{x}_B$ and $\hat{y}_B$

$$\hat{\boldsymbol{x}}_B = \hat{\boldsymbol{x}}_A \cos\theta + \hat{\boldsymbol{y}}_A \sin\theta$$
$$\hat{\boldsymbol{y}}_B = -\hat{\boldsymbol{x}}_A \sin\theta + \hat{\boldsymbol{y}}_A \cos\theta$$

or in matrix form as

$$\begin{pmatrix} \hat{\boldsymbol{x}}_B & \hat{\boldsymbol{y}}_B \end{pmatrix} = \begin{pmatrix} \hat{\boldsymbol{x}}_A & \hat{\boldsymbol{y}}_A \end{pmatrix} \underbrace{\begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}}_{{}^A\mathbf{R}_B(\theta)}$$

where

$${}^A\mathbf{R}_B(\theta) = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$$

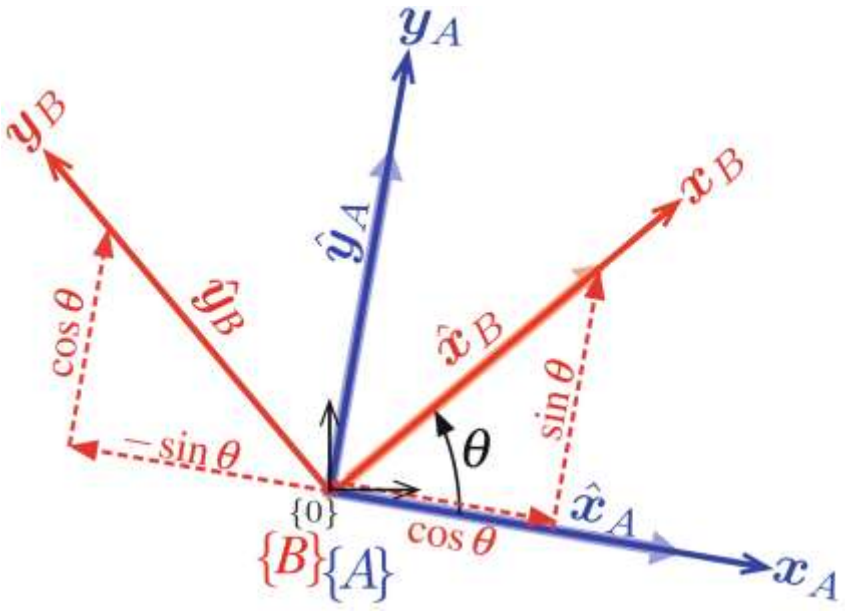


Fig. 2.10 Rotated coordinate frames

Rotation in Two Dimensions (2D)

- Matrix ${}^{A}\mathbf{R}_{B}(\theta)$ is known as a rotation matrix and describes how points are transformed from frame $\{B\}$ to frame $\{A\}$ when the frame is ***rotated*** θ *deg*.
- ${}^{A}\mathbf{R}_{B}$ is orthogonal with the determinant equal to 1 and $\mathbf{R}^{-1} = \mathbf{R}^{T}$ while $\mathbf{R}(-\theta)=\mathbf{R}(\theta)^{T}$
- The matrix-vector product **R**v preserves the length and relative orientation of vectors v
- It is a member of the Special Orthogonal (SO) group of dimension 2 which we write as

$$\mathbf{R} \in \mathbf{SO}(2) \subset \mathbb{R}^{2\times 2}$$

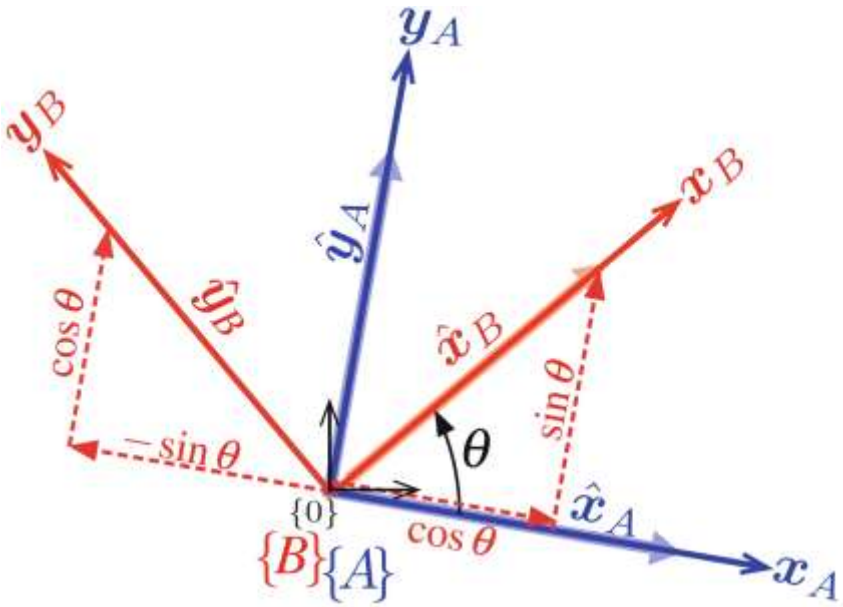


Fig. 2.10 Rotated coordinate frames

2D Rotation Matrix - Properties

- Using a four-element matrix to describe a scalar; rotation angle!
- The properties pose three constraints
  - Both columns has unit magnitude
  - The columns are orthogonal
- Non-minimal representation
  - Increased representation complexity
  - BUT: easy to combine and calculate using normal matrix algebra

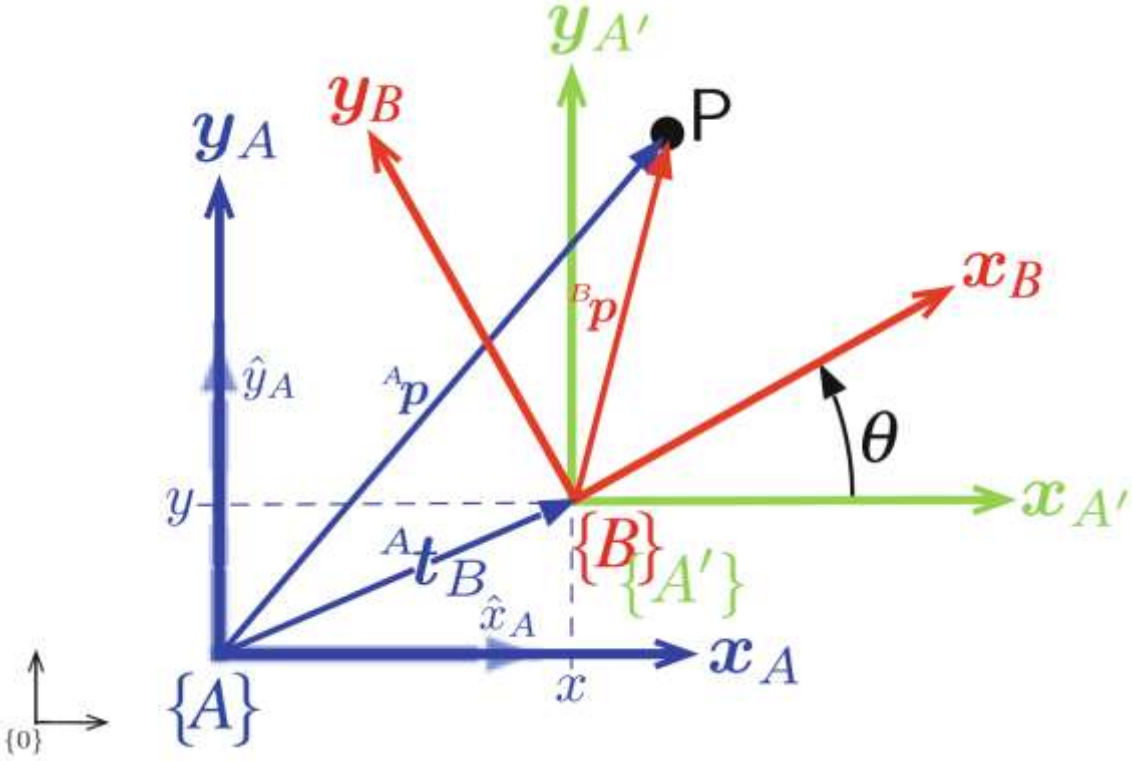
Working in Two Dimensions (2D)

- Translation is simple vectorial addition since {A} and {A'} are parallel

$$\begin{pmatrix} {}^{A}x \\ {}^{A}y \end{pmatrix} = \begin{pmatrix} {}^{A'}x \\ {}^{A'}y \end{pmatrix} + \begin{pmatrix} t_x \\ t_y \end{pmatrix}$$

$$= \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} \begin{pmatrix} {}^{B}x \\ {}^{B}y \end{pmatrix} + \begin{pmatrix} t_x \\ t_y \end{pmatrix}$$

$$= \begin{pmatrix} \cos\theta & -\sin\theta & t_x \\ \sin\theta & \cos\theta & t_y \end{pmatrix} \begin{pmatrix} {}^{B}x \\ {}^{B}y \\ 1 \end{pmatrix}$$



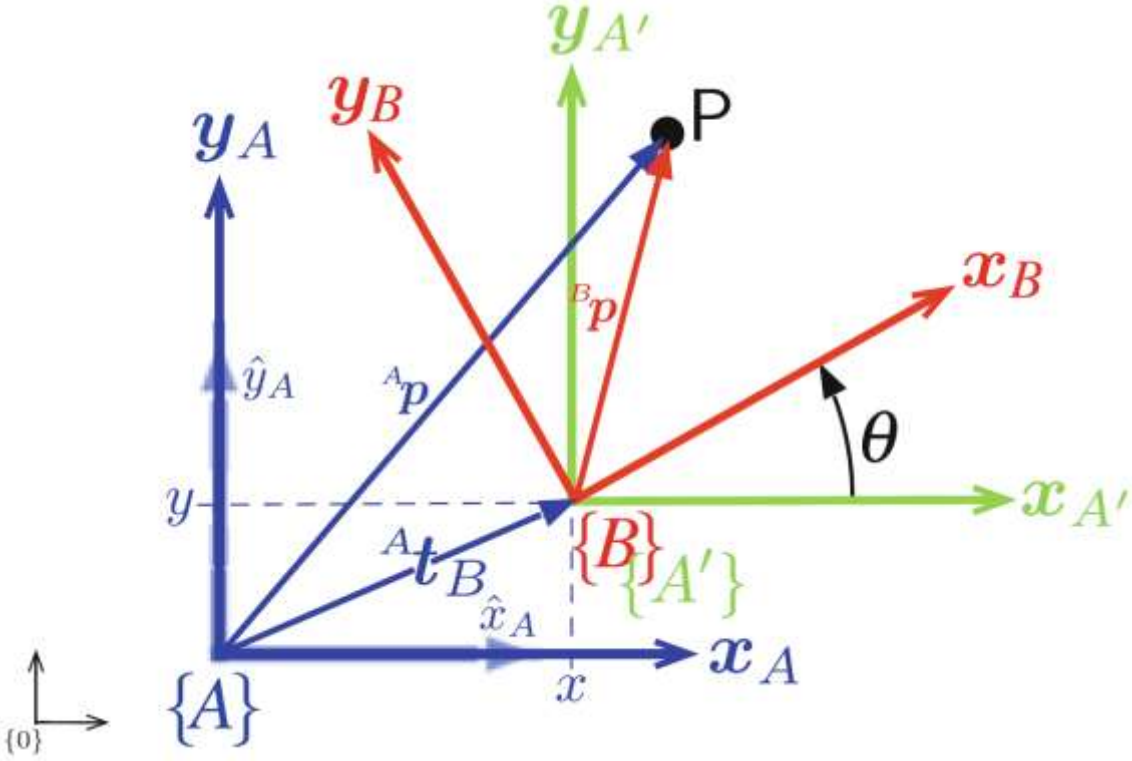
Working in Two Dimensions (2D)

- Or more compactly

$$\begin{pmatrix} {}^{A}x \\ {}^{A}y \\ 1 \end{pmatrix} = \begin{pmatrix} {}^{A}\mathbf{R}_B(\theta) & {}^{A}\boldsymbol{t}_B \\ \mathbf{0}_{1\times 2} & 1 \end{pmatrix} \begin{pmatrix} {}^{B}x \\ {}^{B}y \\ 1 \end{pmatrix}$$

where ${}^{A}\mathbf{t}_B = (t_x, t_y)$ is the translation and ${}^{A}\mathbf{R}_B(\theta)$ is the orientation is of the frame {B} w.r.t. {A}

- Note that ${}^{A}\mathbf{R}_B = {}^{A'}\mathbf{R}_B$ since the axes of frames {A} and {A'} are parallel



Working in Two Dimensions (2D)

- The coordinate vectors for point **P** are now expressed in **homogeneous form** and we write

$$ {}^{A}\tilde{p} = \begin{pmatrix} {}^{A}\mathbf{R}_B(\theta) & {}^{A}\boldsymbol{t}_B \\ \mathbf{0}_{1\times 2} & 1 \end{pmatrix} {}^{B}\tilde{p} = {}^{A}\boldsymbol{T}_B \, {}^{B}\tilde{p} $$

- ${}^{A}\mathbf{T}_B$ represents both translation and orientation or **relative pose** and is referred to as a homogeneous transformation

- It has very specific structure and belongs to the Special Euclidean group of dimension 2:

$$ \mathbf{T} \in \mathbf{SE}(2) \subset \mathbb{R}^{3\times 3} $$

Rotation Translation

$$ T = \begin{pmatrix} \cos\theta & -\sin\theta & x \\ \sin\theta & \cos\theta & y \\ 0 & 0 & 1 \end{pmatrix} $$

Scaling

Working in Two Dimensions (2D)

ξ as an $\mathbf{SE}(2)$ matrix

For the case of rotation and translation in 2D, ξ can be implemented by a homogeneous transformation matrix $\mathbf{T} \in \mathbf{SE}(2)$ which is sometimes written as an ordered pair $(\mathbf{R}, \boldsymbol{t}) \in \mathbf{SO}(2) \times \mathbb{R}^2$. The implementation is:

composition	$\xi_1 \oplus \xi_2$	$\mapsto \mathbf{T}_1\mathbf{T}_2 = \begin{pmatrix} \mathbf{R}_1\mathbf{R}_2 & \boldsymbol{t}_1 + \mathbf{R}_1\boldsymbol{t}_2 \\ \mathbf{0}_{1\times 2} & 1 \end{pmatrix}$, matrix multiplication
inverse	$\ominus\xi$	$\mapsto \mathbf{T}^{-1} = \begin{pmatrix} \mathbf{R}^\top & -\mathbf{R}^\top\boldsymbol{t} \\ \mathbf{0}_{1\times 2} & 1 \end{pmatrix}$, matrix inverse
identity	$\varnothing$	$\mapsto \mathbf{1}_{3\times 3}$, identity matrix
vector-transform	$\xi \cdot \boldsymbol{v}$	$\mapsto \epsilon(\mathbf{T}\tilde{\boldsymbol{v}})$, matrix-vector product

where $\tilde{\cdot} : \mathbb{R}^2 \mapsto \mathbb{P}^2$ and $\epsilon(\cdot) : \mathbb{P}^2 \mapsto \mathbb{R}^2$. Composition is not commutative, that is, $\mathbf{T}_1\mathbf{T}_2 \neq \mathbf{T}_2\mathbf{T}_1$.

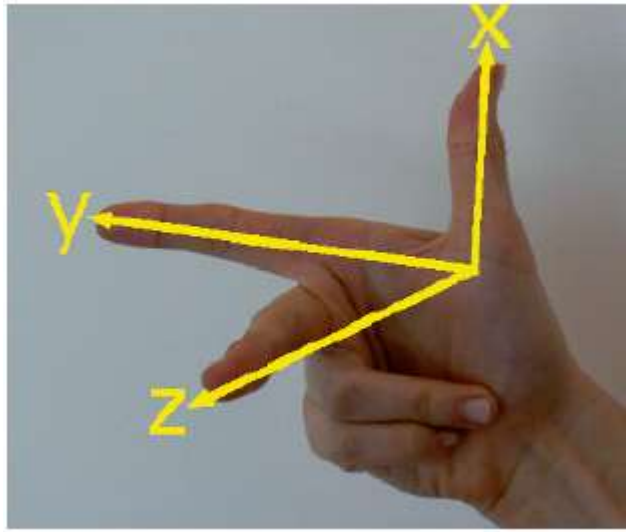
Working in Three Dimensions (3D)

- The ***3-dimensional*** case is an extension of the 2-dimensional case
- The direction of the z-axis obeys the ***right-hand rule*** and forms a ***right-handed coordinate frame***.
- Unit vectors parallel to the axes are denoted $\hat{\boldsymbol{x}}$, $\hat{\boldsymbol{y}}$ and $\hat{\boldsymbol{z}}$ such that

$$\hat{\boldsymbol{z}} = \hat{\boldsymbol{x}} \times \hat{\boldsymbol{y}}, \quad \hat{\boldsymbol{x}} = \hat{\boldsymbol{y}} \times \hat{\boldsymbol{z}}; \quad \hat{\boldsymbol{y}} = \hat{\boldsymbol{z}} \times \hat{\boldsymbol{x}} \qquad (2.20)$$

- A point **P** is represented by its x-, y- and z-coordinates (x, y, z) or as a bound vector

$$\boldsymbol{p} = x\hat{\boldsymbol{x}} + y\hat{\boldsymbol{y}} + z\hat{\boldsymbol{z}}$$



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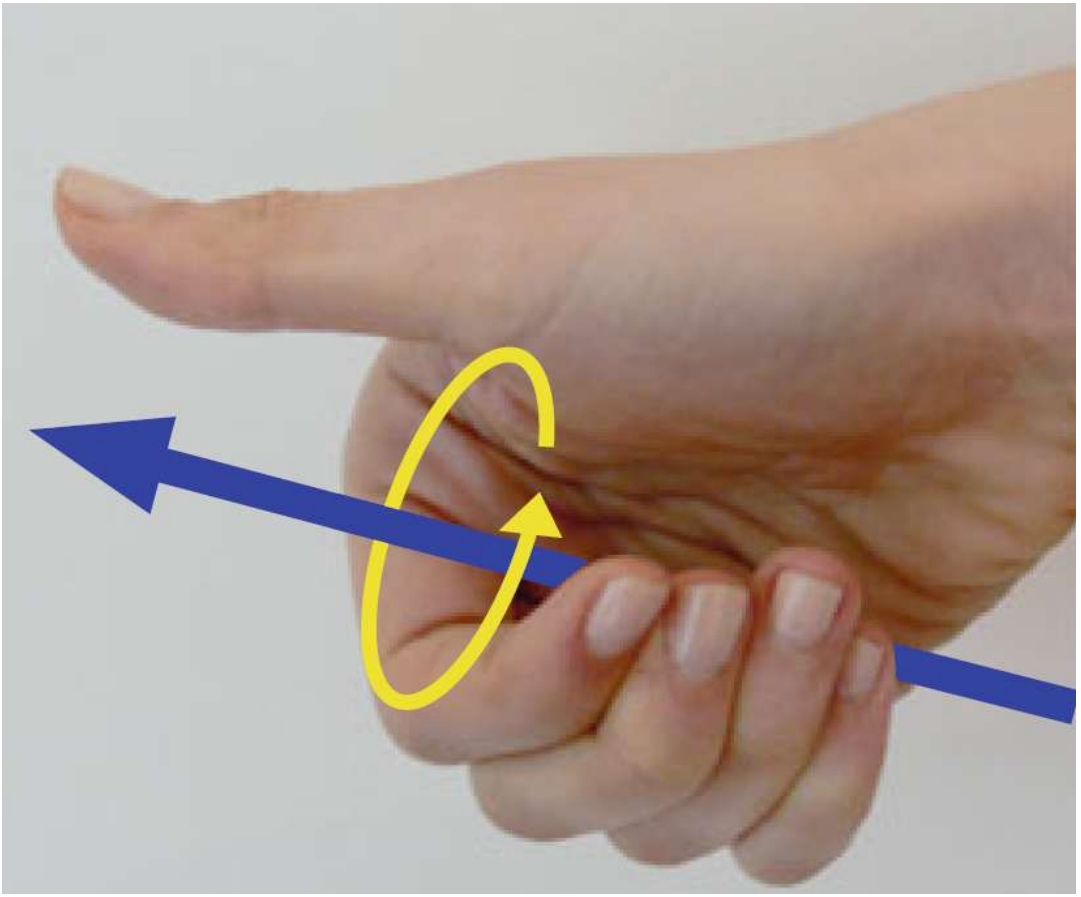
Working in Three Dimensions (3D)

Rotation About a Vector:

- Wrap your right hand around the vector with your thumb (your x-finger) in the direction of the arrow. The curl of your fingers indicates the direction of increasing angle.

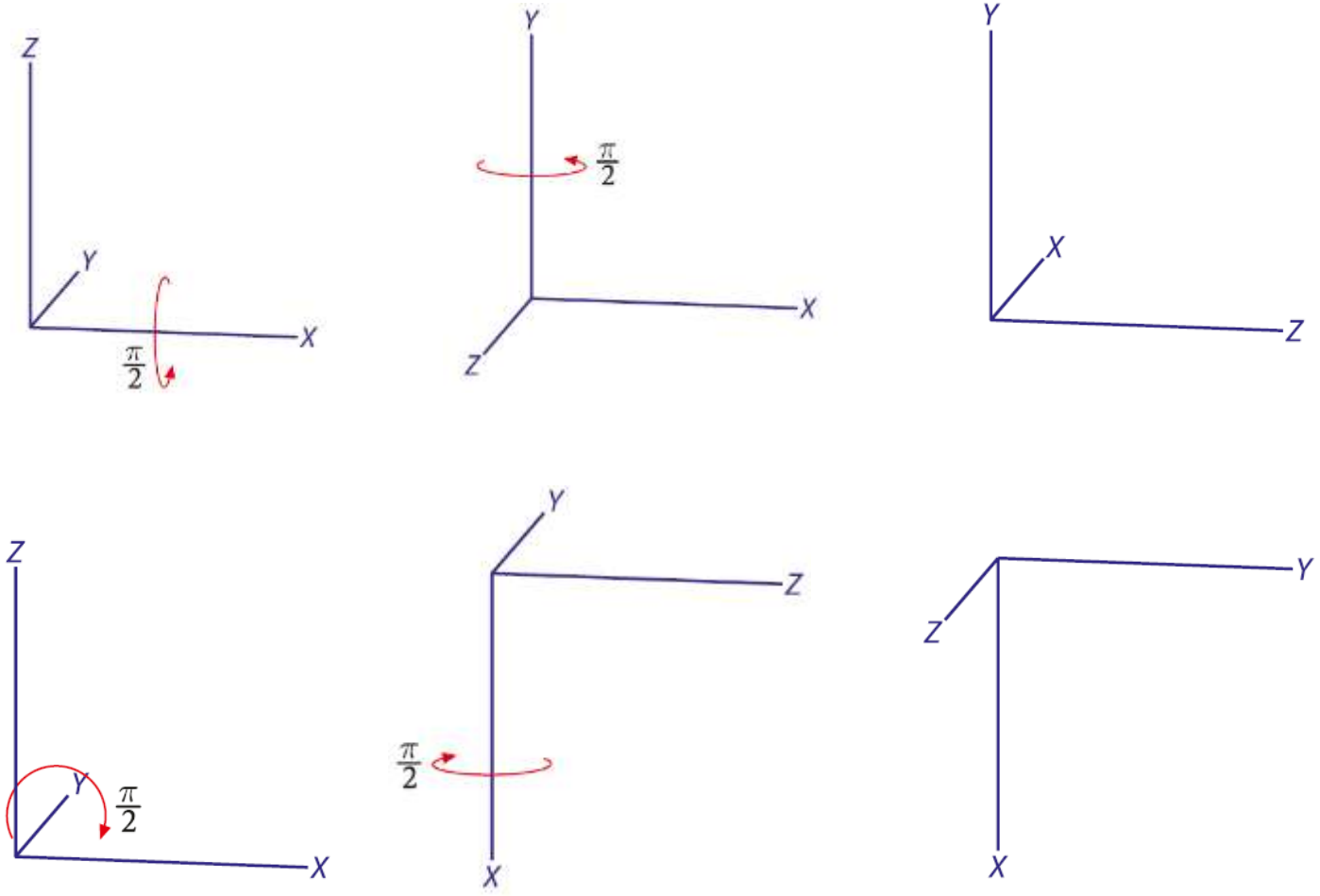


Orientation in 3-Dimensions

- Rotation in 3D is **not** commutative

$$R_x\left(\frac{\pi}{2}\right) R_y\left(\frac{\pi}{2}\right) \neq R_y\left(\frac{\pi}{2}\right) R_x\left(\frac{\pi}{2}\right)$$

- The order in which rotations are applied decides the result

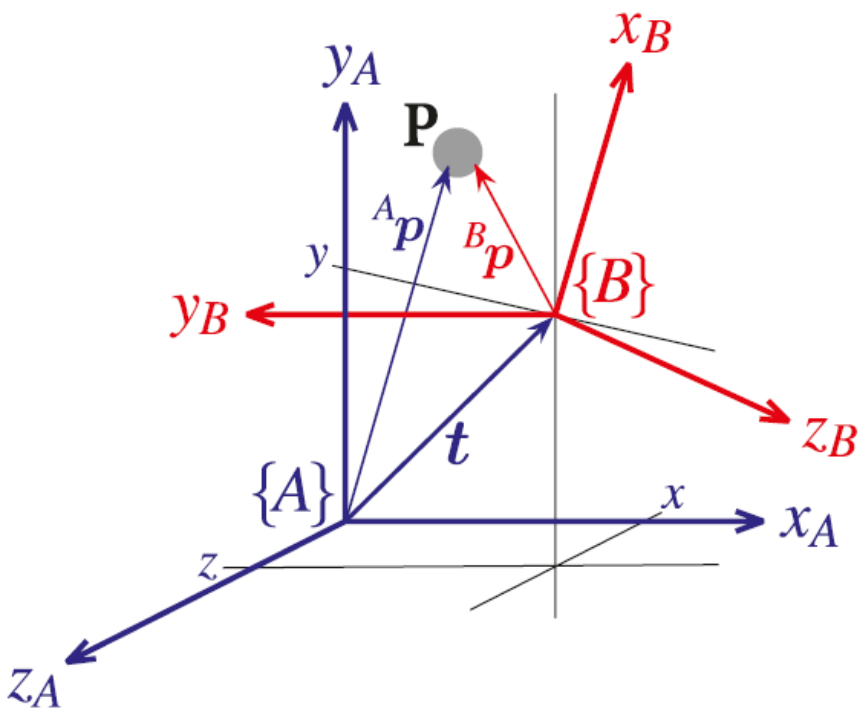


Working in Three Dimensions (3D)

- Figure shows a red coordinate frame $\{B\}$ that we wish to describe with respect to the blue reference frame $\{A\}$.
- We can see that the origin of $\{B\}$ has been displaced by the vector

$$t = (x, y, z)$$

and then rotated in some complex fashion by ${}^{A}R_{B}$.



Euler's rotation theorem states that any rotation can be considered as a sequence of 3 rotations such that no two successive rotations are about the same axis.

Orthonormal Rotation Matrix

- We can represent the ***orientation*** of a coordinate frame by its unit vectors expressed in terms of the reference coordinate frame.
- Each unit vector has three elements, and they form the columns of a 3×3 *orthonormal matrix* ${}^{A}\mathbf{R}_{B}$

$$\begin{pmatrix} {}^{A}p_x \\ {}^{A}p_y \\ {}^{A}p_z \end{pmatrix} = {}^{A}\mathbf{R}_{B} \begin{pmatrix} {}^{B}p_x \\ {}^{B}p_y \\ {}^{B}p_z \end{pmatrix} \tag{2.21}$$

which transforms the description of a vector defined with respect to frame $\{B\}$ to a vector with respect to $\{A\}$.

Orthonormal Rotation Matrix

- The orthonormal rotation matrices for rotation of θ about the x-, y- and z-axes are

$$\mathbf{R}_x(\theta) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta \\ 0 & \sin\theta & \cos\theta \end{pmatrix}$$

$$\mathbf{R}_y(\theta) = \begin{pmatrix} \cos\theta & 0 & \sin\theta \\ 0 & 1 & 0 \\ -\sin\theta & 0 & \cos\theta \end{pmatrix}$$

$$\mathbf{R}_z(\theta) = \begin{pmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Orthonormal Rotation Matrix

ξ as an $\mathbf{SO}(3)$ matrix

For the case of pure rotation in 3D, ξ can be implemented by a rotation matrix $\mathbf{R} \in \mathbf{SO}(3)$. The implementation is:

composition	$\xi_1 \oplus \xi_2$	$\mapsto \mathbf{R}_1\mathbf{R}_2$, matrix multiplication
inverse	$\ominus\xi$	$\mapsto \mathbf{R}^{-1} = \mathbf{R}^\top$, matrix transpose
identity	$\varnothing$	$\mapsto \mathbf{1}_{3\times3}$, identity matrix
vector-transform	$\xi \cdot \boldsymbol{v}$	$\mapsto \mathbf{R}\boldsymbol{v}$, matrix-vector product

Composition is not commutative, that is, $\mathbf{R}_1\mathbf{R}_2 \neq \mathbf{R}_2\mathbf{R}_1$.

Three-Angle Representation

- ***Euler's rotation theorem*** requires successive rotation about three axes such that no two successive rotations are about the same axis.
- **"Euler angles"** is an ambiguous term

- The ***ZYZ*** angle sequence $\mathbf{R}(\phi, \theta, \psi) = \mathbf{R}_z(\phi)\,\mathbf{R}_y(\theta)\,\mathbf{R}_z(\psi)$

is commonly used in aeronautics and mechanical dynamics

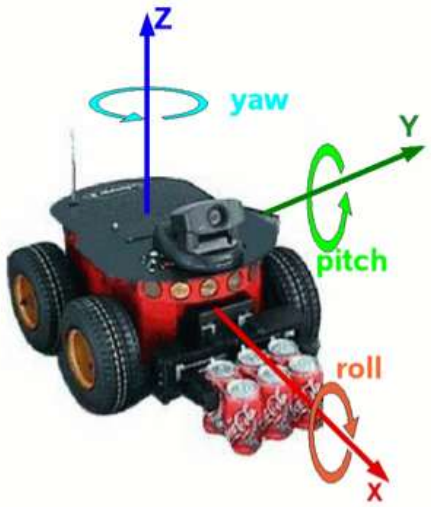
- The Euler angles are the 3-vector: $\boldsymbol{\Gamma} = (\phi, \theta, \psi) \in (\mathbf{S}^1)^3 \text{ or } \mathbf{S}^1 \times \mathbf{S}^1 \times \mathbf{S}^1$

Three-Angle Representation

- There are a total of twelve unique rotation sequences.
  - XYX, XZX, YXY, YZY, ZXZ, ZYZ (rotation about one particular axis)
  - XYZ, XZY, YZX, YXZ, ZXY, ZYX (roation about all three axes)
- The particular angle sequence is convention within a particular technology field

Roll-pitch-yaw *angles are common for describing vehicle attitudes, e.g., aircrafts, ships and cars*

- There are two different roll-pitch-yaw sequences in common use:
  - ZYX for mobile robots and XYZ for robot manipulators
- The ***ZYX*** angle sequence $\mathbf{R}(\alpha, \beta, \gamma) = \mathbf{R}_z(\gamma)\,\mathbf{R}_y(\beta)\,\mathbf{R}_x(\alpha)$

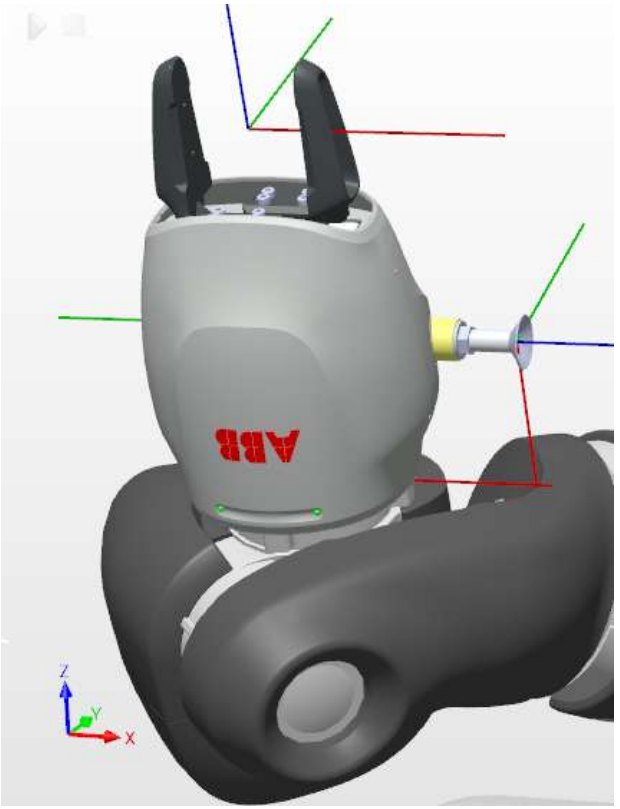


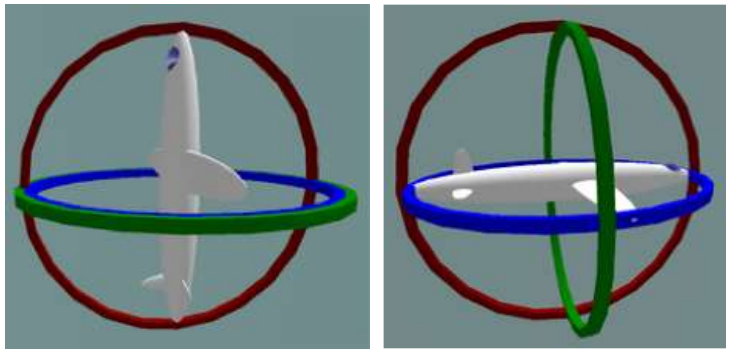
Three-Angle Representation

- When describing the attitude of a robot gripper the convention is that the z-axis points forward and the x-axis is either up or down.
- This leads to the ***XYZ*** angle sequence

$$\mathbf{R}(\alpha, \beta, \gamma) = \mathbf{R}_x(\gamma)\,\mathbf{R}_y(\beta)\,\mathbf{R}_z(\alpha) \tag{2.24}$$

- A fundamental problem with all the three-angle representations is ***singularity***. This is also known as gimbal lock.
- This occurs when the ***rotational axis*** of the ***middle*** term in the sequence becomes parallel to the rotation axis of the ***first*** or ***third*** term – effectively loosing 1 Degree of Freedom (DoF)





Quarternions: non-minimal representation

- Extension of complex numbers

$$\boldsymbol{q} = s + \boldsymbol{v} = s + v_1 i + v_2 j + v_3 k$$

- Where $s \in \mathbb{R}$, $\boldsymbol{v} \in \mathbb{R}^3$ and

$$i^2 = j^2 = k^2 = ijk = -1$$

- Denoted $\mathbf{q} = s < v_1, v_2, v_3 >$
- Multiplication (Hamilton Product):

$$q_1.q_2 = s_1 s_2 - v_1.v_2 < s_1 v_2 + s_2 v_1 + v_1 \times v_2 >$$

- Quaternions are non-commutiative
- Rotations are represented by non-singular **unit quaternions**

Norm: $|q|2 = s^2 + {v_1}^2 + {v_2}^2 + {v_3}^2$

Conjugate quaternion: $\bar{q} = \boldsymbol{s - v_1 i - v_2 j - v_3 k}$

Inverse quaternion: $q^{-1} = \bar{q}/|q|^2$

Unit quaternion: $|q|=1$

Inverse Unit quaternion: $|q|^{-1}=1$

$$\breve{q} \circ \breve{q}' = \begin{pmatrix} s & -v_x & -v_y & -v_z \\ v_x & s & v_z & -v_y \\ v_y & v_z & s & v_x \\ v_z & v_y & -v_x & s \end{pmatrix} \begin{pmatrix} s' \\ v'_x \\ v'_y \\ v'_z \end{pmatrix}$$

Quarternions: non-minimal representation

ξ as a unit quaternion

For the case of pure rotation in 3D, ξ can be implemented with a unit quaternion $\mathring{q} \in S^3$. The implementation is:

composition	$\xi_1 \oplus \xi_2$	$\mapsto \mathring{q}_1 \circ \mathring{q}_2$, Hamilton product
inverse	$\ominus \xi$	$\mapsto \mathring{q}^* = s\langle -\boldsymbol{v} \rangle$, quaternion conjugation
identity	$\varnothing$	$\mapsto 1\langle 0 \rangle$
vector-transform	$\xi \cdot \boldsymbol{v}$	$\mapsto \mathring{q} \circ \check{v} \circ \mathring{q}^*$, where $\check{v} = 0\langle \boldsymbol{v} \rangle$ is a pure quaternion

Composition is not commutative, that is, $\mathring{q}_1 \circ \mathring{q}_2 \neq \mathring{q}_2 \circ \mathring{q}_1$.

Pose in 3-Dimensions

- The derivation for the homogeneous transformation matrix is given by:

$$\begin{pmatrix} {}^{A}x \\ {}^{A}y \\ {}^{A}z \\ 1 \end{pmatrix} = \begin{pmatrix} {}^{A}\mathbf{R}_{B} & {}^{A}\boldsymbol{t}_{B} \\ \mathbf{0}_{1\times 3} & 1 \end{pmatrix} \begin{pmatrix} {}^{B}x \\ {}^{B}y \\ {}^{B}z \\ 1 \end{pmatrix}$$

where ${}^{\mathrm{A}}\boldsymbol{t}_{\mathrm{B}} \in \mathbb{R}^3$ is a vector defining the ***origin*** of frame $\{\mathrm{B}\}$ with respect to frame $\{A\}$, and ${}^{\mathrm{A}}\boldsymbol{R}_{\mathrm{B}} \in SO(3)$ is the 3×3 orthonormal matrix which describes the orientation of the axes of frame $\{B\}$ with respect to frame $\{A\}$

Pose in 3-Dimensions

- If points are represented by homogeneous coordinate vectors, then

$$
\begin{aligned}
{}^{A}\tilde{\boldsymbol{p}} &= \begin{pmatrix} {}^{A}\mathbf{R}_{B} & {}^{A}\boldsymbol{t}_{B} \\ \mathbf{0}_{1\times 3} & 1 \end{pmatrix} {}^{B}\tilde{\boldsymbol{p}} \\
&= {}^{A}\mathbf{T}_{B}\, {}^{B}\tilde{\boldsymbol{p}}
\end{aligned}
\tag{2.36}
$$

Where ${}^{A}\mathbf{T}_{B}$ is a 4×4 homogeneous transformation matrix

- **T** belongs to the Special Euclidean group of dim 3: $\mathbf{T} \in \mathbf{SE}(3) \subset \mathbb{R}^{4x4}$

- The 4×4 homogeneous transformation is ***very commonly*** used in robotics, computer graphics and computer vision.

Pose in 3-Dimensions: Properties

ξ as an SE(3) matrix

For the case of rotation and translation in 3D, ξ can be implemented by a homogeneous transformation matrix $\mathbf{T} \in \mathbf{SE}(3)$ which is sometimes written as an ordered pair $(\mathbf{R}, \boldsymbol{t}) \in \mathbf{SO}(3) \times \mathbb{R}^3$. The implementation is:

composition	$\xi_1 \oplus \xi_2$	$\mapsto \mathbf{T}_1\mathbf{T}_2 = \begin{pmatrix} \mathbf{R}_1\mathbf{R}_2 & \boldsymbol{t}_1 + \mathbf{R}_1\boldsymbol{t}_2 \\ \mathbf{0}_{1\times 3} & 1 \end{pmatrix}$, matrix multiplication
inverse	$\ominus\xi$	$\mapsto \mathbf{T}^{-1} = \begin{pmatrix} \mathbf{R}^\top & -\mathbf{R}^\top\boldsymbol{t} \\ \mathbf{0}_{1\times 3} & 1 \end{pmatrix}$, matrix inverse
identity	$\varnothing$	$\mapsto \mathbf{1}_{4\times 4}$, identity matrix
vector-transform	$\xi \cdot \boldsymbol{v}$	$\mapsto \epsilon(\mathbf{T}\tilde{\boldsymbol{v}})$, matrix-vector product using homogeneous vectors

where $\tilde{\cdot} : \mathbb{R}^3 \mapsto \mathbb{P}^3$ and $\epsilon(\cdot) : \mathbb{P}^3 \mapsto \mathbb{R}^3$. Composition is not commutative, that is, $\mathbf{T}_1\mathbf{T}_2 \neq \mathbf{T}_2\mathbf{T}_1$.

Configuration space (Ch. 2.4.9)

- So far, we have considered the pose of objects; position & orientation in 2D/3D
- An alternative way of thinking about this comes from classical mechanics and is referred to as the **configuration** of a system.
- The configuration is the smallest set of parameters, called generalized coordinates, that are required to fully describe the position of *every* particle in the system.
- The number of independent generalized coordinates, *N,* is known as the number of Degrees of Freedom (DoF) of the system
- Any configuration of the system is represented by a point in its *N*-dimensional configuration space, or C-space, denoted by C and $\mathbf{q} \in C$. We have dim(C) = N.
- Over/under actuated: number of control inputs are larger/smaller than N.
- All under-actuated systems are also non-holonomic (will get back to this in Ch. 4)

Mobile Robots – Locomotion and configuration space examples

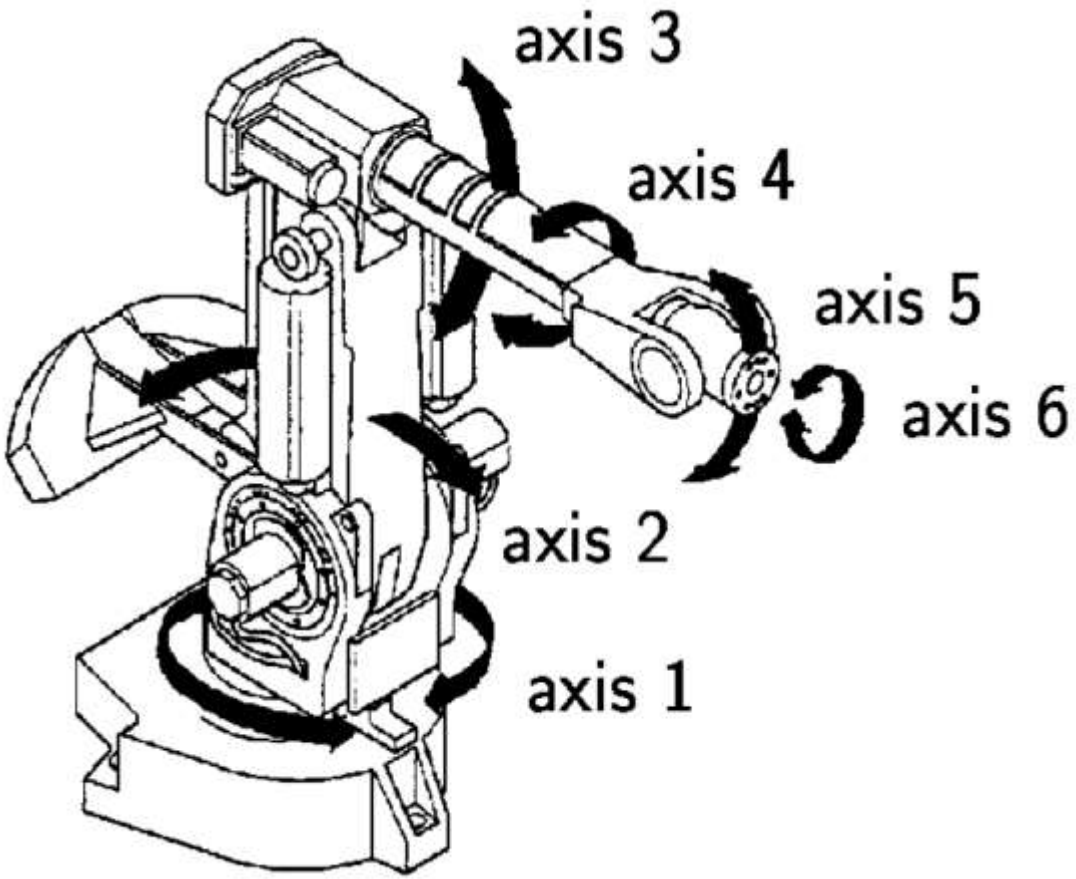
Train	Hover	Helicopter	Fixed-wing aircraft	Quadrotor	Car
1 - DoF $q = x$	3 - DoF $\boldsymbol{q} = (x, y, \vartheta_z)$	6 - DoF $\boldsymbol{q} = (x, y, z, \vartheta_r, \vartheta_p, \vartheta_y)$	6 - DoF $\boldsymbol{q} = (x, y, z, \vartheta_r, \vartheta_p, \vartheta_y)$	6 - DoF $\boldsymbol{q} = (x, y, z, \vartheta_r, \vartheta_p, \vartheta_y)$	3 - DoF $\boldsymbol{q} = (x, y, \vartheta_z)$

Configuration space of industrial robots

- Most traditional industrial robots have 6-axis (DoF) and 6 actuators

  $q = (q_1, q_2, q_3, q_4, q_5, q_6)$

- The C-space and task-space have the same dimension (holonomic system)



Configuration space of ABB YuMi

- ABB YuMi has 7 axis (over-actuated). This to improve reachability & manipulability
- Still, singularities exist as depicted in [this video]

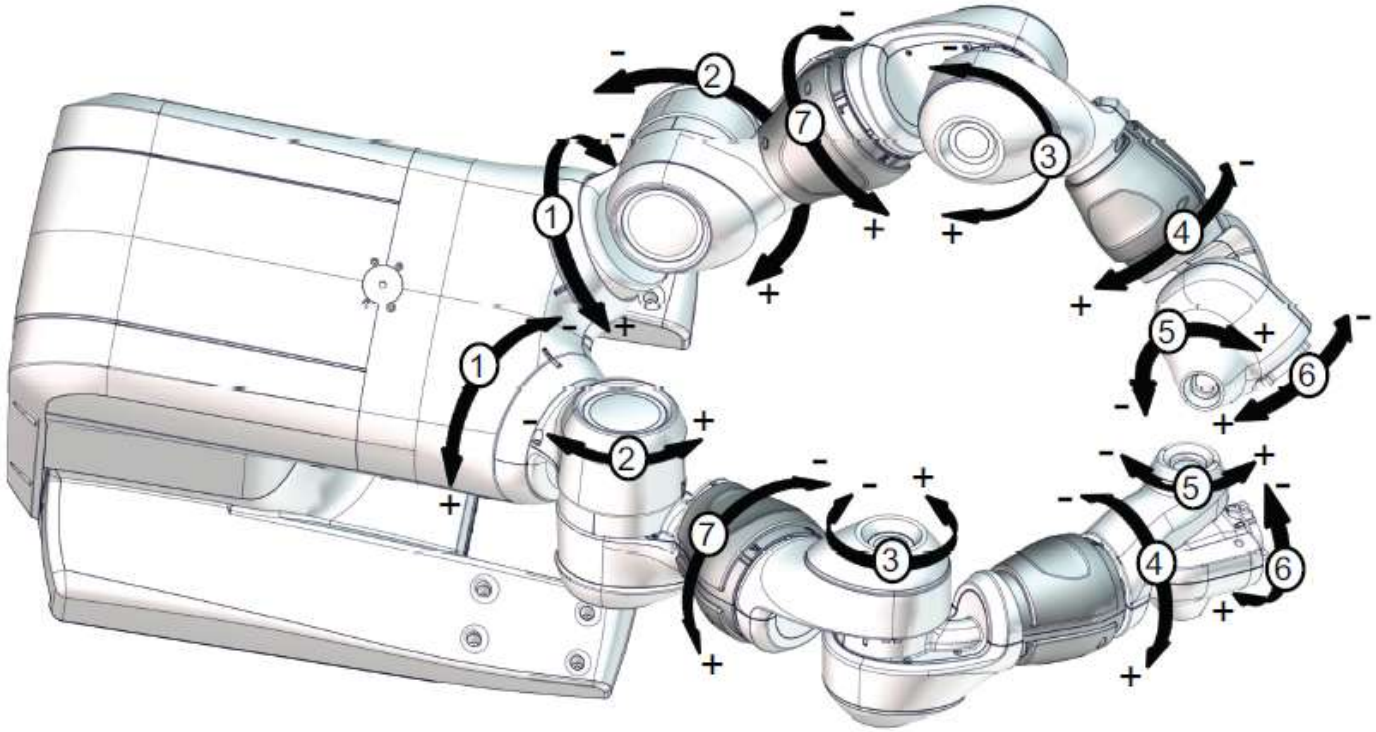


Photo courtesy: ABB Robotics

Summary – some takes aways from Ch 2.

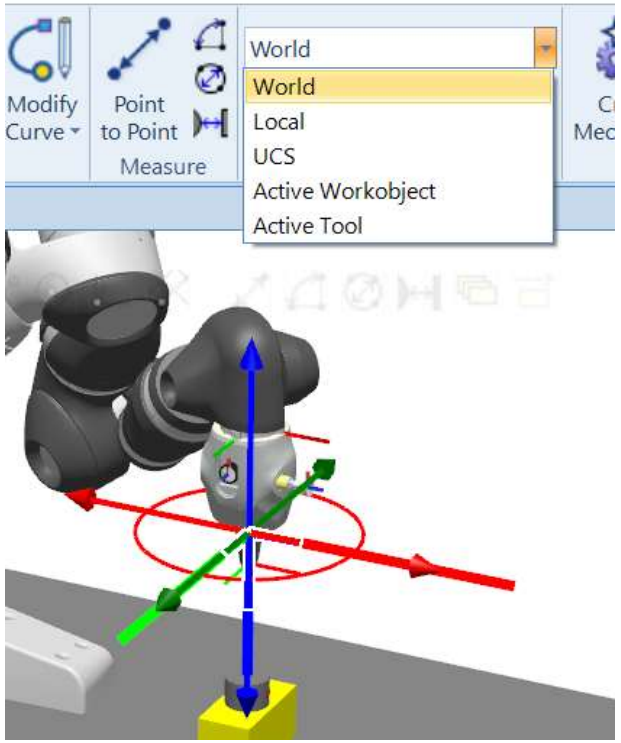
Lesson 1: There are many mathematical objects that can represent relative pose.

- No right or wrong; all have their pros and cons; but all 3 angular descriptions have singularities

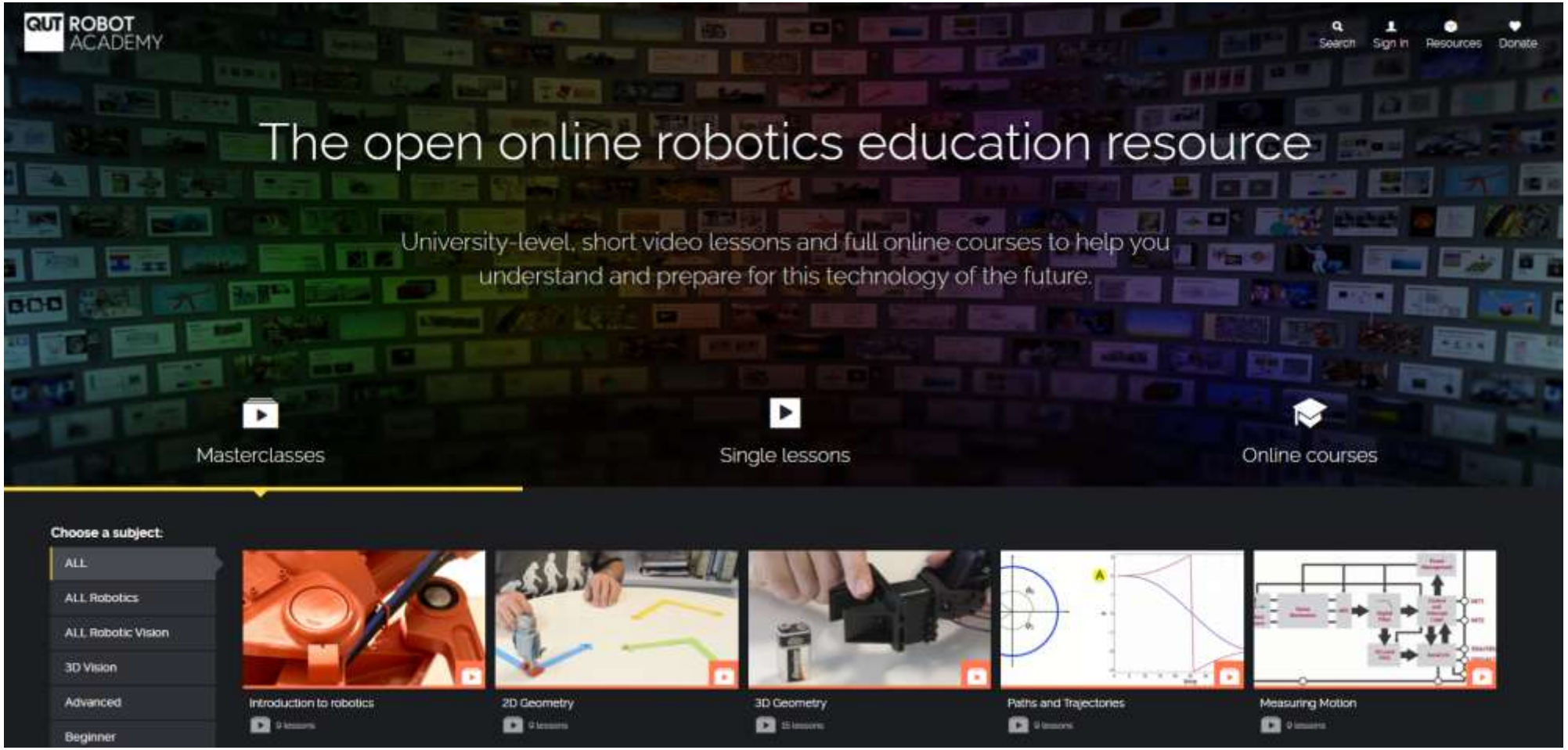
Lesson 2: coordinate frame are your friend, embrace them!

- Start by assigning coordinate frames to all objects of interest

Lesson 3: robots can be described in both task space (2D/3D) and configuration/joint space. We will utilize this in the upcoming lab!



Online videos: QUT Robot Academy



Thank you for your attention !